

Simple Harmonic Motion

Question1

If the function $\sin^2 \omega t$ (t is time in second) represents a periodic motion, then the period of the motion is

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Options:

A.

$$\sqrt{\frac{\pi}{\omega}} \text{ s}$$

B.

$$\frac{\pi}{\omega} \text{ s}$$

C.

$$\frac{2\pi}{\omega} \text{ s}$$

D.

$$\sqrt{\frac{2\pi}{\omega}} \text{ s}$$

Answer: B

Solution:

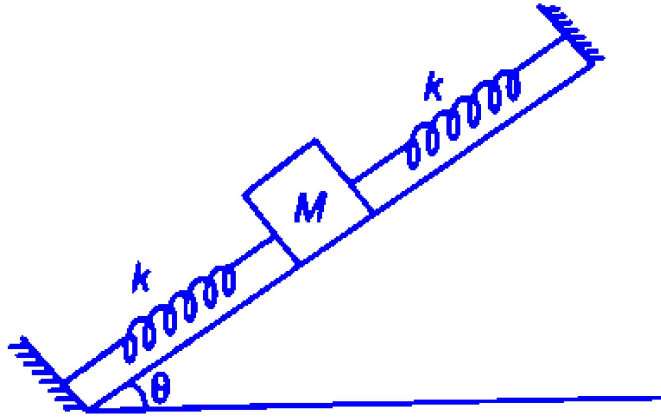
$$\sin^2 \omega t = \frac{1 - \cos 2\omega t}{2} = \frac{1}{2} - \frac{1}{2} \cos 2\omega t$$

$$\therefore T = \frac{2\pi}{2\omega} = \frac{\pi}{\omega} \text{ s}$$

Question2

On a smooth inclined plane a block of mass M is fixed to two rigid supports using two springs as shown in the figure. If each spring has spring constant k , then the period of oscillation of the block is

(Neglect the masses of the springs)



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Options:

A.

$$2\pi \left(\frac{M}{2k} \right)^{1/2}$$

B.

$$2\pi \left(\frac{2M}{k} \right)^{1/2}$$

C.

$$2\pi \left(\frac{Mg \sin \theta}{2k} \right)^{1/2}$$

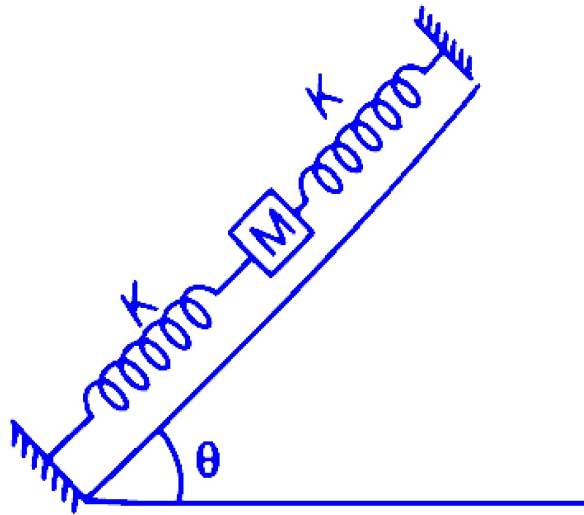
D.

$$2\pi \left(\frac{2Mg}{k} \right)^{1/2}$$

Answer: A

Solution:

According to given figure, springs are connected in parallel.



Therefore effective spring constant

$$\begin{aligned} K_{\text{eff}} &= K_1 + K_2 \\ &= K + K = 2K \end{aligned}$$

The restoring force on block due to springs not due to gravity pull

$$\text{So, } T = 2\pi\sqrt{\frac{M}{K_{\text{eff}}}} \Rightarrow T = 2\pi\sqrt{\frac{M}{2K}}$$

Question3

A particle is executing simple harmonic motion with amplitude A . The ratio of the kinetic energies of the particle when it is at displacements of $\frac{A}{4}$ and $\frac{A}{2}$ from the mean position is

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Options:

A.

4 : 1

B.

2 : 1

C.

5 : 4

D.

9 : 16

Answer: C

Solution:

$$\begin{aligned}\frac{k_1}{k_2} &= \frac{\frac{1}{2}m\omega^2 (A^2 - x_1^2)}{\frac{1}{2}m\omega^2 (A^2 - x_2^2)} \\ &= \frac{A^2 - x_1^2}{A^2 - x_2^2} = \frac{A^2 - \left(\frac{A}{4}\right)^2}{A^2 - \left(\frac{A}{2}\right)^2} \\ &= \frac{\frac{15}{16}A^2}{\frac{3}{4}A^2} = \frac{5}{4}\end{aligned}$$

Question4

A particle is executing simple harmonic motion starting from its mean position. If the time period of the particle is 1.5 s , then the minimum time at which the ratio of the kinetic and total energies of the particle becomes 3:4 is

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Options:

A.

$$\frac{1}{4} \text{ s}$$

B.

$$\frac{1}{12} \text{ s}$$

C.

$$\frac{1}{8} \text{ s}$$

D.

$$\frac{1}{6} \text{ s}$$

Answer: C

Solution:

For particle in SHM,

$$T = 1.5 \text{ s}$$

$$\omega = \frac{2\pi}{T} = \frac{4\pi}{3} \text{ rad/s}$$

Let oscillations starts from $t = 0$ and $x = A \sin \omega t$ is displacement equation Then, KE at time t is,

$$\begin{aligned} K &= \frac{1}{2}mv^2 = \frac{1}{2}m\left(\frac{dx}{dt}\right)^2 \\ &= \frac{1}{2}m \cdot A^2\omega^2 \cos^2 \omega t \quad \dots (i) \end{aligned}$$

And total energy of particle is,

$$E = \frac{1}{2}kA^2$$

Here, $k = m\omega^2$

$$= \frac{1}{2}mA^2\omega^2 \quad \dots (ii)$$

$$\begin{aligned} \text{Given, ratio, } \frac{K}{E} &= \frac{3}{4} \\ \Rightarrow \frac{\frac{1}{2}mA^2\omega^2 \cos^2 \omega t}{\frac{1}{2}mA^2\omega^2} &= \frac{3}{4} \\ \Rightarrow \cos^2 \omega t &= \frac{3}{4} \\ \Rightarrow \cos \omega t &= \sqrt{\frac{3}{4}} \\ \Rightarrow \omega t &= \frac{\pi}{6} \\ \Rightarrow t &= \frac{\pi}{6\omega} = \frac{\pi \times 3}{6 \times 4\pi} = \frac{1}{8} \text{ s} \end{aligned}$$

Question5

The equations for the displacements of two particles in simple harmonic motion are $y_1 = 0.1 \sin \left(100\pi t + \frac{\pi}{3}\right)$ and $y_2 = 0.1 \cos \pi t$ respectively. The phase difference between the velocities of the two particles at a time $t = 0$ is

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Options:

A.

$$\frac{\pi}{4}$$

B.

$$\frac{\pi}{2}$$

C.

$$\frac{\pi}{6}$$

D.

$$\frac{\pi}{3}$$

Answer: C

Solution:

$$y_1 = 0.1 \sin \left(100\pi t + \frac{\pi}{3} \right)$$

$$\begin{aligned} \therefore v_1 &= 0.1 \times 100\pi \cos \left(100\pi t + \frac{\pi}{3} \right) \\ &= 10\pi \times \cos \left(100\pi t + \frac{\pi}{3} \right) \end{aligned}$$

$$y_2 = 0.1 \cos \pi t$$

$$= 0.1 \sin \left(\pi t + \frac{\pi}{2} \right)$$

$$\begin{aligned} \therefore v_2 &= 0.1 \times \pi \cos \left(\pi t + \frac{\pi}{2} \right) \\ &= 0.1\pi \cos \left(\pi t + \frac{\pi}{2} \right) \end{aligned}$$

$\therefore$ At $t = 0$

$$v_1 = 10 \cos \frac{\pi}{3}$$

$$\text{and } v_2 = 0.1\pi \cos \frac{\pi}{2}$$

$\therefore$ Phase difference between v_1 and v_2 at time $t = 0$,

is given as,

$$\phi = \frac{\pi}{2} - \frac{\pi}{3} = \frac{\pi}{6}$$

Question6

A spring is stretched by 0.2 m when a mass of 0.5 kg is suspended to it. The time period of the spring when 0.5 kg mass is replaced with a mass of 0.25 kg is suspended to it is

(Acceleration due to gravity = 10 ms^{-2})

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Options:

A.

0.628 s

B.

6.28 s

C.

62.8 s

D.

0.0628 s

Answer: A

Solution:

Step 1: Find Spring Constant (k)

The spring stretches by 0.2 m when a 0.5 kg mass hangs from it.

This means the force from the weight (m_1g) equals the force from the spring (kx):

$$kx = m_1g$$

So,

$$k = \frac{m_1g}{x} = \frac{0.5 \times 10}{0.2} = 25 \text{ N/m}$$

Step 2: Use the New Mass

The new hanging mass is $m_2 = 0.25 \text{ kg}$.

Step 3: Find the New Time Period (T)

The time period of a mass-spring system is:

$$T = 2\pi\sqrt{\frac{m_2}{k}}$$

Substitute the values:

$$T = 2\pi\sqrt{\frac{0.25}{25}}$$

$$T = 2\pi\sqrt{0.01}$$

$$T = 2\pi \times 0.1 = 0.2\pi \text{ s}$$

Step 4: Calculate the Value

Using $\pi \approx 3.14$:

$$T = 0.2 \times 3.14 = 0.628 \text{ s}$$

Question7

A body of mass 4 kg attached to a spring of force constant 64Nm^{-1} executes simple harmonic motion on a frictionless horizontal surface. The time period of oscillation is

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Options:

A.

$$\frac{\pi}{3} \text{ s}$$

B.

$$\frac{\pi}{2} \text{ s}$$

C.

$$\pi \text{ s}$$

D.

$$\frac{3\pi}{2} \text{ s}$$

Answer: B

Solution:

- Mass $m = 4 \text{ kg}$
- Force constant (spring constant) $k = 64 \text{ N/m}$

Formula for time period of a mass–spring system:

$$T = 2\pi\sqrt{\frac{m}{k}}$$

Substitute the values:

$$T = 2\pi\sqrt{\frac{4}{64}} = 2\pi\sqrt{\frac{1}{16}} = 2\pi \times \frac{1}{4} = \frac{\pi}{2}$$

 **Answer:** $\boxed{\frac{\pi}{2} \text{ s}}$

Correct Option: B

Question8

A particle is executing simple harmonic motion with amplitude A . At a distance ' x ' from the mean position, when the particle is moving towards extreme position it receives a blow in the direction of motion which instantaneously doubles its velocity. The new amplitude of the particle is

(Frequency is constant during the motion)

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Options:

A.

A

B.

$$\sqrt{A^2 - X^2}$$

C.

$$\sqrt{2A^2 - 3x^2}$$

D.

$$\sqrt{4A^2 - 3x^2}$$

Answer: D

Solution:

Let angular frequency of SHM is ω and when is at distance x from mean position, its velocity will be given as

$$v = \omega\sqrt{A^2 - x^2} \quad \dots (i)$$

When its velocity becomes double, let new amplitude is A' .

$$\text{So, } 2v = \omega\sqrt{A'^2 - x^2}$$

Putting the value of v from equation (i)

$$\begin{aligned} 2\omega\sqrt{A^2 - x^2} &= \omega\sqrt{A'^2 - x^2} \\ 2\sqrt{A^2 - x^2} &= \sqrt{A'^2 - x^2} \\ \Rightarrow 4(A^2 - x^2) &= A'^2 - x^2 \\ \Rightarrow A'^2 &= 4A^2 - 3x^2 \Rightarrow A' = \sqrt{4A^2 - 3x^2} \end{aligned}$$

Question9

If the displacement ' x ' of a body in motion in terms of time ' t ' is given by $x = A \sin(\omega t + \theta)$, then the minimum time at which the displacement becomes maximum is

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Options:

A.

$$\left[\frac{\pi}{2\omega} - \frac{\theta}{\omega} \right]$$

B.

$$\left[\frac{2\omega}{\pi} - \frac{\omega}{\theta} \right]$$

C.

$$\left[\frac{\pi}{\omega} - \frac{1}{\omega} \right]$$

D.

$$\left[\frac{\omega}{\pi} - \frac{\omega}{\pi^2} \right]$$

Answer: A

Solution:

$$x = A \sin(\omega t + \theta)$$

For maximum displacement, $x = A$

$$A = A \sin(\omega t + \theta)$$

$$\Rightarrow 1 = \sin(\omega t + \theta)$$

$$\Rightarrow \omega t + \theta = \frac{\pi}{2}$$

$$\Rightarrow \omega t = \frac{\pi}{2} - \theta$$

$$\Rightarrow t = \frac{\pi}{2\omega} - \frac{\theta}{\omega}$$

Question10

If the maximum velocity and maximum acceleration of a particle executing simple harmonic motion are respectively 5 ms^{-1} and 10 ms^{-2} , then the time period of oscillation of the particle is

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Options:

A.

$$\pi \text{ s}$$

B.

$$2\pi \text{ s}$$

C.

2 s

D.

1 s

Answer: A

Solution:

Given:

- Maximum velocity $v_{\max} = 5 \text{ m/s}$
- Maximum acceleration $a_{\max} = 10 \text{ m/s}^2$

For SHM relationships:

$$v_{\max} = \omega A$$

$$a_{\max} = \omega^2 A$$

where ω is the angular frequency, and A is the amplitude.

Step 1: Divide the two equations

$$\frac{a_{\max}}{v_{\max}} = \frac{\omega^2 A}{\omega A} = \omega$$

$$\Rightarrow \omega = \frac{a_{\max}}{v_{\max}} = \frac{10}{5} = 2 \text{ rad/s}$$

Step 2: Find the time period

$$T = \frac{2\pi}{\omega} = \frac{2\pi}{2} = \pi \text{ s}$$



Final Answer:

$$T = \pi \text{ s}$$

Correct Option: A) $\pi \text{ s}$

Question11

A body of mass 1 kg is suspended from a spring of force constant 600 Nm^{-1} . Another body of mass 0.5 kg moving vertically upwards

hits the suspended body with a velocity of 3 ms^{-1} and embedded in it. The amplitude of motion is

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Options:

A.

5 cm

B.

15 cm

C.

10 cm

D.

8 cm

Answer: A

Solution:

Combined mass, $m = m_1 + m_2$

$$= 1 + 0.5 = 1.5 \text{ kg}$$

Using conservation of momentum,

$$m_1 u_1 + m_2 u_2 = (m_1 + m_2) v$$

$$\Rightarrow 1 \times 0 + 0.5 \times 3 = 1.5 v$$

$$\Rightarrow v = 1 \text{ m/s}$$

According to conservation of energy,

$$\frac{1}{2} m v^2 = \frac{1}{2} k A^2$$

$$\Rightarrow A = \sqrt{\frac{m v^2}{k}}$$

$$\begin{aligned} \Rightarrow \sqrt{\frac{1.5 \times 1^2}{600}} &= \sqrt{\frac{1}{400}} = \frac{1}{20} \text{ m} \\ &= \frac{100}{20} \text{ cm} = 5 \text{ cm} \end{aligned}$$

Question12

For a particle executing simple harmonic motion, the ratio of kinetic and potential energies at a point where displacement is one half of the amplitude is

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Options:

A.

3 : 1

B.

1 : 3

C.

2 : 1

D.

1 : 2

Answer: A

Solution:

$$\begin{aligned}\frac{K}{U} &= \frac{\frac{1}{2}m\omega^2 (A^2 - x^2)}{\frac{1}{2}m\omega^2 x^2} \\ &= \frac{A^2 - x^2}{x^2} \\ &= \frac{A^2 - \left(\frac{A}{2}\right)^2}{\left(\frac{A}{2}\right)^2} \left[\text{Here, } x = \frac{A}{2} \right] \\ &= \frac{\frac{3A^2}{4}}{\frac{A^2}{4}} = \frac{3}{1}\end{aligned}$$

$$\therefore K : U = 3 : 1$$

Question13

When the mass attached to a spring is increased from 4 kg to 9 kg , the time period of oscillation increases by 0.2π s. Then, the spring constant of the spring is

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Options:

A.

$$80 \text{ N} - \text{m}^{-1}$$

B.

$$200 \text{ N} - \text{m}^{-1}$$

C.

$$50 \text{ N} - \text{m}^{-1}$$

D.

$$100 \text{ N} - \text{m}^{-1}$$

Answer: D

Solution:

$$T = 2\pi\sqrt{\frac{m}{k}}$$

$$\frac{T_1}{T_2} = \sqrt{\frac{m_1}{m_2}}$$

$$\frac{T}{T + 0.2\pi} = \sqrt{\frac{4}{9}} = \frac{2}{3}$$

$$\Rightarrow 3T = 2T + 0.4\pi$$

$$\Rightarrow T = 0.4\pi \text{ s}$$

$\therefore$ Using Eq. (i),

$$T = 2\pi\sqrt{\frac{m}{k}}$$

$$\Rightarrow 0.4\pi = 2\pi\sqrt{\frac{4}{k}} \Rightarrow 0.2 = \frac{2}{\sqrt{k}}$$

$$\Rightarrow \sqrt{k} = \frac{1}{0.1} \Rightarrow \sqrt{k} = 10$$

$$\Rightarrow k = 100\text{Nm}^{-1}$$

Question14

The kinetic energy of a particle executing simple harmonic motion at a displacement of 3 cm from the mean position is 4 mJ . If the amplitude of the particle is 5 cm , then the maximum force acting on the particle is

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Options:

A.

0.25 N

B.

0.50 N

C.

0.75 N

D.

1.25 N

Answer: A

Solution:

Total energy,

$$E = \frac{1}{2}m\omega^2 A^2$$

$$K + U = \frac{1}{2}m\omega^2(0.05)^2$$

$$\Rightarrow 4 \times 10^{-3} + \frac{1}{2}m\omega^2 x^2 = \frac{1}{2}m\omega^2(0.05)^2$$

$$\Rightarrow \frac{1}{2}m\omega^2 [(0.05)^2 - x^2] = 4 \times 10^{-3}$$

$$\Rightarrow \frac{1}{2}m\omega^2 [(0.05)^2 - (0.03)^2] = 4 \times 10^{-3}$$

$$\Rightarrow \frac{1}{2}m\omega^2 = 2.5$$

$$\Rightarrow m\omega^2 = 5$$

$\therefore$ Maximum force

$$F_{\max} = m\omega^2 A = 5 \times 0.05 = 0.25 \text{ N}$$

Question15

A body of mass 1 kg is attached to the lower end of a vertically suspended spring of force constant $600 \text{ N} - \text{m}^{-1}$. If another body of mass 0.5 kg moving vertically upward hits the suspended body with a velocity 3 ms^{-1} and embedded in it, then the frequency of the oscillation is

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Options:

A.

$$\frac{5}{\pi} \text{ Hz}$$

B.

$$\frac{10}{\pi} \text{ Hz}$$

C.

$$\frac{\pi}{5} \text{ Hz}$$

D.

$$\pi \text{ Hz}$$

Answer: B

Solution:

Angular velocity of the system,

$$\omega = \sqrt{\frac{k}{m_{\text{total}}}} = \sqrt{\frac{600}{(1 + 0.5)}}$$
$$= \sqrt{\frac{600}{1.5}} = 20 \text{ rad/s}$$

$$\therefore f = \frac{\omega}{2\pi} = \frac{20}{2\pi} = \frac{10}{\pi} \text{ Hz}$$

Question16

If the displacement y (in cm) of a particle executing simple harmonic motion is given by the equation

$y = 5 \sin(3\pi t) + 5\sqrt{3} \cos(3\pi t)$, then the amplitude of the particle is

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Options:

A.

5 cm

B.

$5(1 + \sqrt{3})\text{cm}$

C.

$5\sqrt{3} \text{ cm}$

D.

10 cm

Answer: D

Solution:

$$\begin{aligned}
 y &= 5 \sin(3\pi t) + 5\sqrt{3} \cos(3\pi t) \\
 &= 10 \left[\frac{5}{10} \sin 3\pi t + \frac{5\sqrt{3}}{10} \cos(3\pi t) \right] \\
 &= 10 \left[\sin 3\pi t \cdot \left(\frac{1}{2} \right) + \cos 3\pi t \cdot \left(\frac{\sqrt{3}}{2} \right) \right] \\
 &= 10 \left[\sin(3\pi t) \cos \frac{\pi}{3} + \cos 3\pi t \sin \frac{\pi}{3} \right] \\
 &= 10 \sin \left(3\pi t + \frac{\pi}{3} \right) \\
 \therefore A &= 10 \text{ cm}
 \end{aligned}$$

Question17

The angular frequency of a block of mass 0.1 kg oscillating with the help of a spring of force constant $2.5 \text{ N} - \text{m}^{-1}$ is

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Options:

A.

$$0.2 \text{ rad s}^{-1}$$

B.

$$5 \text{ rad s}^{-1}$$

C.

$$10 \text{ rad s}^{-1}$$

D.

$$2 \text{ rad s}^{-1}$$

Answer: B

Solution:

$$n = \frac{1}{2\pi} \sqrt{\frac{k}{m}} = \frac{1}{2\pi} \sqrt{\frac{25}{0.1}}$$

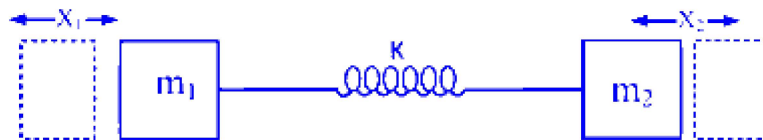
$$n = \frac{5}{2\pi} \text{ Hz}$$

$\therefore$ Angular frequency, $\omega = 2\pi n$

$$= 2\pi \times \frac{5}{2\pi} = 5 \text{ rad s}^{-1}$$

Question18

As shown in the figure, two blocks of masses m_1 and m_2 are connected to spring of force constant k . The blocks are slightly displaced in opposite directions to x_1, x_2 distances and released. If the system executes simple harmonic motion, then the frequency of oscillation of the system (ω) is



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Options:

- A. $\left(\frac{1}{m_1} + \frac{1}{m_2} \right) k^2$
- B. $\sqrt{\left(\frac{1}{m_1} + \frac{1}{m_2} \right) k^2}$
- C. $\sqrt{\left(\frac{1}{m_1} + \frac{1}{m_2} \right)}$
- D. $\sqrt{\left(\frac{1}{m_1} + \frac{1}{m_2} \right) k}$

Answer: A

Solution:

To determine the frequency of oscillation for the system, we need to consider the concept of reduced mass, which is useful when analyzing systems involving two masses connected by a spring.

First, the reduced mass (μ) of a two-body system is given by the following formula:

$$\frac{1}{\mu} = \frac{1}{m_1} + \frac{1}{m_2}$$

where m_1 and m_2 are the masses of the two blocks connected to the spring. The reduced mass essentially simplifies the system to behave as if it were a single object of mass μ .

Once the reduced mass is calculated, the frequency of oscillation (ω) for a spring-mass system is determined by the formula:

$$\omega = \sqrt{\frac{k}{\mu}}$$

Substituting the expression for reduced mass into the formula, we get:

$$\omega = \sqrt{k \left(\frac{1}{m_1} + \frac{1}{m_2} \right)}$$

This expression gives the angular frequency of the simple harmonic motion for the system, where k is the spring constant. This encapsulates how the two masses and the spring constant interact to influence the oscillation frequency.

Question19

A mass M , attached to a horizontal spring executes simple harmonic motion with amplitude A_1 . When mass M passes mean position, then a smaller mass m is attached to it and both of them together executing simple harmonic motion with amplitude A_2 . Then, value of $\frac{A_1}{A_2}$ is

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Options:

A. $\sqrt{\frac{m^2+M^2}{M^2}}$

B. $\sqrt{\frac{m+M}{M^2}}$

C. $\sqrt{\frac{m+M}{M}}$

D. $\frac{m+M}{M}$

Answer: C

Solution:

To find the ratio of amplitudes, $\frac{A_1}{A_2}$, for the simple harmonic motion (SHM) described, we need to consider the changes when a mass m is added to a mass M , which is attached to a spring.

Key Equations:

Time Period for Mass $(M + m)$:

$$T_2 = 2\pi\sqrt{\frac{m+M}{k}}$$

Time Period for Mass M Only:

$$T_1 = 2\pi\sqrt{\frac{M}{k}}$$

Velocity Considerations:

Assume v_1 is the velocity of the mass M at the mean position and v_2 is the velocity of the combined mass $(m + M)$.

Using Conservation of Linear Momentum:

$$Mv_1 = (m + M)v_2$$

Since $v_1 = A_1\omega_1$ and $v_2 = A_2\omega_2$:

$$M(A_1\omega_1) = (m + M)(A_2\omega_2)$$

Rewriting for the amplitude ratio $\frac{A_1}{A_2}$:

$$\frac{A_1}{A_2} = \frac{(m+M)}{M} \times \frac{\omega_2}{\omega_1}$$

Given $\omega_1 = \frac{2\pi}{T_1}$ and $\omega_2 = \frac{2\pi}{T_2}$, we have:

$$\frac{\omega_2}{\omega_1} = \frac{T_1}{T_2}$$

Substituting Time Periods:

Substituting from earlier equations:

$$\frac{T_1}{T_2} = \sqrt{\frac{M}{m+M}}$$

Therefore:

$$\frac{A_1}{A_2} = \frac{(m+M)}{M} \times \sqrt{\frac{M}{m+M}}$$

$$\frac{A_1}{A_2} = \sqrt{\frac{m+M}{M}}$$

This provides the required ratio of the amplitudes when a smaller mass is attached, leading to a new simple harmonic motion with the described properties.

Question20

The displacement of a particle of mass 2 g executing simple harmonic motion is $x = 8 \cos \left(50t + \frac{\pi}{12} \right)$ m, where t is time in second. The maximum kinetic energy of the particle is

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Options:

A. 160 J

B. 80 J

C. 40 J

D. 20 J

Answer: A

Solution:

The particle has a mass of:

$$m = 2 \text{ g} = 2 \times 10^{-3} \text{ kg}$$

The displacement of the particle is described by:

$$x = 8 \cos \left(50t + \frac{\pi}{12} \right) \text{ m}$$

The maximum velocity $v_{\max}$ can be determined using the formula:

$$v_{\max} = A\omega$$

where $A = 8 \text{ m}$ and $\omega = 50 \text{ rad/s}$. Thus:

$$v_{\max} = 8 \times 50 = 400 \text{ m/s}$$

The maximum kinetic energy ($\text{KE}_{\max}$) is given by:

$$\text{KE}_{\max} = \frac{1}{2} m (v_{\max})^2$$

Substituting the known values:

$$\begin{aligned} \text{KE}_{\max} &= \frac{1}{2} \times 2 \times 10^{-3} \times (4 \times 10^2)^2 \\ &= 10^{-3} \times 16 \times 10^4 \end{aligned}$$

Therefore, the maximum kinetic energy is:

$$KE_{\max} = 160 \text{ J}$$

Question21

The relation between the force (F in Newton) acting on a particle executing simple harmonic motion and the displacement of the particle (y in metre) is $500F + \pi^2 y = 0$. If the mass of the particle is 2 g . The time period of oscillation of the particle is

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Options:

A. 8 s

B. 6 s

C. 2 s

D. 4 s

Answer: C

Solution:

Given:

Mass of the particle: $m = 2 \text{ g} = 2 \times 10^{-3} \text{ kg}$

Displacement-force relationship: $500F + \pi^2 y = 0$

From this equation, we can express the force F in terms of the displacement y :

$$F = \frac{-\pi^2 y}{500}$$

In simple harmonic motion (SHM), force is also given by:

$$F = -ky$$

By comparing the two force equations, we identify the spring constant k :

$$k = \frac{\pi^2}{500}$$

The angular frequency ω for SHM is determined by:

$$\omega = \sqrt{\frac{k}{m}}$$

Substituting the values for k and m :

$$\omega = \sqrt{\frac{\pi^2}{500 \times 2 \times 10^{-3}}} = \pi \text{ rad/s}$$

The time period T of SHM is given by:

$$T = \frac{2\pi}{\omega}$$

Plugging in the value of ω :

$$T = \frac{2\pi}{\pi} \Rightarrow T = 2 \text{ seconds}$$

Therefore, the time period of oscillation of the particle is 2 seconds.

Question22

Two simple harmonic motions are represented by $y_1 = 5[\sin 2\pi t + \sqrt{3} \cos 2\pi t]$ and $y_2 = 5 \sin \left[2\pi t + \frac{\pi}{4}\right]$. The ratio of their amplitudes is

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Options:

A. 1 : 1

B. 2 : 1

C. 1 : 3

D. $\sqrt{3}$: 1

Answer: B

Solution:

Here, given

$$y_1 = 5[\sin 2\pi t + \sqrt{3} \cos 2\pi t]$$

$$y_2 = 5 \sin \left[2\pi t + \frac{\pi}{4} \right]$$

$$\begin{aligned} \text{For } y_1 &= 10 \left[\frac{1}{2} \sin 2\pi t + \frac{\sqrt{3}}{2} \cos 2\pi t \right] \\ &= 10 \left(\cos \frac{\pi}{3} \sin 2\pi t + \sin \frac{\pi}{3} \cos 2\pi t \right) \\ y_1 &= 10 \left[\sin 2\pi t + \frac{\pi}{3} \right] \end{aligned}$$

Amplitude in y_1 is $A_1 = 10$

Similarly, $y_2 = 5 \sin \left(2\pi t + \frac{\pi}{4} \right)$

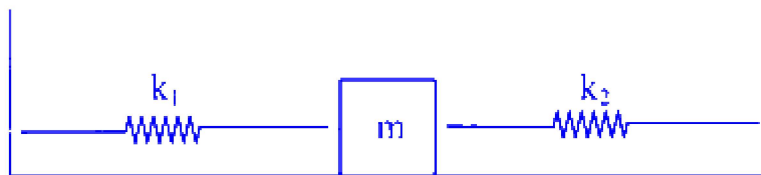
Amplitude in y_2 is $A_2 = 5$

Thus, $\frac{A_1}{A_2} = \frac{10}{5} = \frac{2}{1}$

The ratio of their amplitude 2 : 1.

Question23

When a mass m is connected individually to the springs k_1 and k_2 , the oscillation frequencies are v_1 and v_2 . If the same mass is attached to the two springs as shown in the figure, the oscillation frequency would be



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Options:

A. $v_1 + v_2$

B. $\sqrt{v_1^2 + v_2^2}$

C. $\left(\frac{1}{v_1} + \frac{1}{v_2}\right)^{-1}$

D. $\sqrt{v_1^2 + v_2^2}$

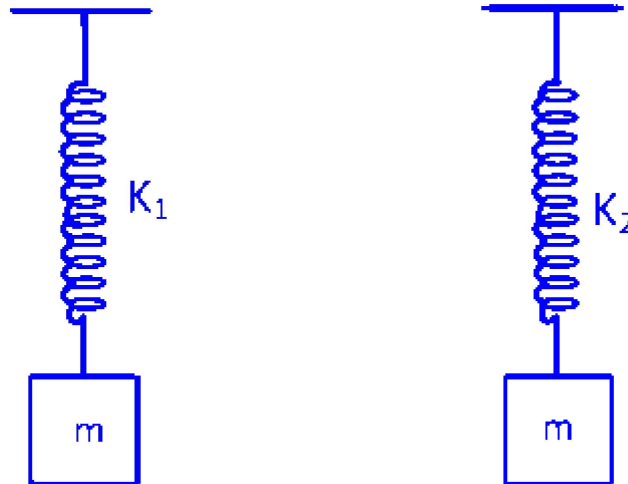
Answer: B

Solution:

When the mass is connected to the two spring individually.

$$v_1 = \frac{1}{2\pi} \sqrt{\frac{k_1}{m}} \quad \dots (i)$$

$$v_2 = \frac{1}{2\pi} \sqrt{\frac{k_2}{m}} \quad \dots (ii)$$



Now, the block is connected with two springs, $k_{eq} = k_1 + k_2$

$$\text{Time period, } T = 2\pi \sqrt{\frac{m}{k_{eq}}}$$

$$= 2\pi \sqrt{\frac{m}{k_1 + k_2}}$$

$$\text{Frequency} = \frac{1}{T} = \frac{1}{2\pi} \times \sqrt{\frac{k_1 + k_2}{m}} \quad \dots (iii)$$

$$v = \frac{1}{2\pi} \left[\frac{k_1}{m} + \frac{k_2}{m} \right]^{1/2}$$

$$\text{From Eq. (i)} \quad \frac{k_1}{m} = 4\pi^2 v_1^2 \text{ and } \frac{k_2}{m} = 4\pi^2 v_2^2$$

$$v = \frac{1}{2\pi} \left[\frac{4\pi^2 v_1^2}{1} + \frac{4\pi^2 v_2^2}{1} \right]^{1/2}$$

$$= \frac{2\pi}{2\pi} [v_1^2 + v_2^2]^{1/2}$$

$$v = \sqrt{v_1^2 + v_2^2}$$

Question24

One bar magnet is in simple harmonic motion with time period T in an earth's magnetic field. If its mass is increased by 9 times the time period becomes

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Options:

A. $3T$

B. $9T$

C. $4T$

D. $\sqrt{3}T$

Answer: A

Solution:

We start with the formula for the time period of a magnetic dipole in a magnetic field:

$$T = 2\pi\sqrt{\frac{I}{MB}} \quad \dots(i)$$

where I is the moment of inertia, M is the magnetic moment, and B is the magnetic field.

Since the moment of inertia I is directly proportional to the mass m of the magnet—meaning $I \propto m$ —if the mass is increased by 9 times, then:

$$I' = 9I$$

The new time period T' can be expressed as:

$$T' = 2\pi\sqrt{\frac{9I}{MB}} \quad \dots(ii)$$

By dividing equation (ii) by equation (i), we find:

$$\frac{T'}{T} = \frac{2\pi\sqrt{\frac{9I}{MB}}}{2\pi\sqrt{\frac{I}{MB}}}$$

Simplifying gives:

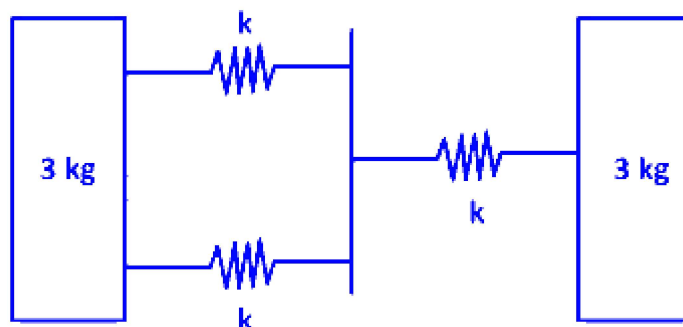
$$\frac{T'}{T} = \sqrt{9} = 3$$

Therefore, the new time period T' is three times the original time period:

$$T' = 3T$$

Question25

In a spring block system as shown in figure. If the spring constant $k = 9\pi^2 \text{Nm}^{-1}$, then the time period of oscillation is



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Options:

- A. 1 s
- B. 3.14 s
- C. 1.414 s
- D. 0.5 s

Answer: A

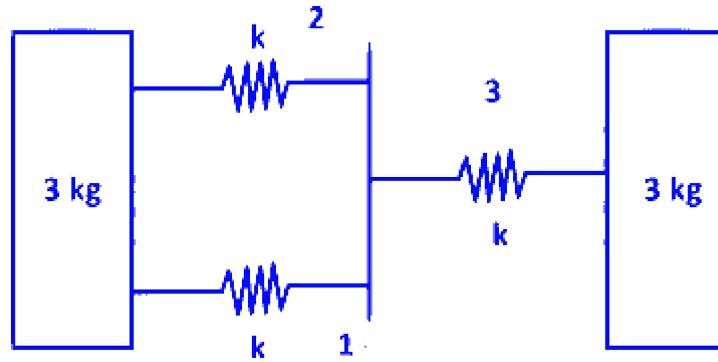
Solution:

Given, $k = 9\pi^2 \text{ N/m}$

Reduced mass of system is

$$\mu = \frac{m_1 m_2}{m_1 + m_2}$$

$$\Rightarrow \mu = \frac{3 \times 3}{3 + 3} = \frac{3}{2} \text{ kg} \quad \dots (i)$$



Now, parallel combination of spring constant (between (1) and (2))

$$k_{\text{eq}} = k + k = 2k$$

Now, k_{eq} and (3) are in series connection,

$$\begin{aligned} k'_{\text{eq}} &= \frac{2k \times k}{2k + k} = \frac{2}{3}k \quad \dots (ii) \\ &= \frac{2}{3} \times 9\pi^2 \\ k'_{\text{eq}} &= 6\pi^2 \end{aligned}$$

Time period is given by formula,

$$\begin{aligned} T &= 2\pi \sqrt{\frac{\mu}{k}} \\ &= 2\pi \sqrt{\frac{3}{2} \times \frac{1}{6\pi^2}} = 1 \text{ s} \end{aligned}$$

Question26

A body is executing simple harmonic motion. At a displacement x its potential energy is E_1 and at a displacement y its potential energy is E_2 . The potential energy E at a displacement $(x + y)$ is

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Options:

A. $\sqrt{E} = \sqrt{E_1} - \sqrt{E_2}$

B. $\sqrt{E} = \sqrt{E_1} + \sqrt{E_2}$

C. $E = E_1 - E_2$

D. $E = E_1 + E_2$

Answer: B

Solution:

Given the scenario where a body is undergoing simple harmonic motion:

The potential energy at displacement x is denoted as E_1 .

Since the body is in simple harmonic motion, the relation is given by:

$$E_1 = \frac{1}{2}m\omega^2 x^2$$

From this, we can derive:

$$\sqrt{E_1} = x\sqrt{\frac{1}{2}m\omega^2} \quad \dots(i)$$

At displacement y , the potential energy is E_2 .

Similarly:

$$E_2 = \frac{1}{2}m\omega^2 y^2$$

Leading to:

$$\sqrt{E_2} = y\sqrt{\frac{1}{2}m\omega^2} \quad \dots(ii)$$

For displacement $(x + y)$, the potential energy is denoted by E .

It can be expressed as:

$$E = \frac{1}{2}m\omega^2 (x + y)^2$$

Therefore:

$$\sqrt{E} = (x + y)\sqrt{\frac{1}{2}m\omega^2} = x\sqrt{\frac{1}{2}m\omega^2} + y\sqrt{\frac{1}{2}m\omega^2}$$

By using equations (i) and (ii), we find:

$$\sqrt{E} = \sqrt{E_1} + \sqrt{E_2}$$

Question27

The mass of a particle is 1 kg and it is moving along X -axis. The period of its oscillation is $\frac{\pi}{2}$. Its potential energy at a displacement of 0.2 m is

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Options:

A. 0.24 J

B. 0.48 J

C. 0.32 J

D. 0.16 J

Answer: C

Solution:

Given:

Mass, $m = 1$ kg

Displacement, $x = 0.2$ m

Period, $T = \frac{\pi}{2}$

We need to find the potential energy at this displacement.

First, we recall the formula for the period of oscillation:

$$T = \frac{2\pi}{\omega}$$

The angular frequency ω is related to the mass m and the spring constant k by:

$$\omega = \sqrt{\frac{k}{m}}$$

Substituting this expression for ω into the period equation, we have:

$$T = \frac{2\pi \cdot \sqrt{m}}{\sqrt{k}}$$

Given $T = \frac{\pi}{2}$, we can solve for k :

$$\frac{\pi}{2} = \frac{2\pi \times \sqrt{1}}{\sqrt{k}}$$

Solving for k :

$$\frac{\pi}{2} = \frac{2\pi}{\sqrt{k}}$$

$$\sqrt{k} = 4$$

$$k = 16 \text{ N/m}$$

Now, compute the potential energy U at the displacement $x = 0.2$ m using:

$$U = \frac{1}{2} kx^2$$

Substitute the values:

$$U = \frac{1}{2} \times 16 \times (0.2)^2$$

$$U = 8 \times 0.04$$

$$U = 0.32 \text{ J}$$

Therefore, the potential energy at a displacement of 0.2 m is 0.32 J.

Question 28

The potential energy of a particle of mass 10 g as a function of displacement x is $(50x^2 + 100)$ J. The frequency of oscillation is

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Options:

A. $\frac{10}{\pi} \text{ s}^{-1}$

B. $\frac{5}{\pi} \text{ s}^{-1}$

C. $\frac{100}{\pi} \text{ s}^{-1}$

D. $\frac{50}{\pi} \text{ s}^{-1}$

Answer: D

Solution:

Given a particle with mass $m = 10 \text{ g} = 0.01 \text{ kg}$, the potential energy as a function of displacement x is expressed as:

$$U(x) = 50x^2 + 100$$

We recognize that the potential energy for a harmonic oscillator is given by:

$$U(x) = \frac{1}{2}kx^2$$

To find the spring constant k , we compare the two expressions:

$$\frac{1}{2}kx^2 = 50x^2$$

Solving for k :

$$\frac{1}{2}k = 50 \Rightarrow k = 100$$

The frequency of oscillation f is derived from the angular frequency ω , given by:

$$f = \frac{\omega}{2\pi} = \frac{1}{2\pi} \cdot \sqrt{\frac{k}{m}}$$

Substituting the values for k and m :

$$f = \frac{1}{2\pi} \cdot \sqrt{\frac{100}{0.01}} f = \frac{100}{2\pi} f = \frac{50}{\pi} \text{ s}^{-1}$$

Question29

A horizontal board is performing simple harmonic oscillations horizontally with an amplitude 0.3 m and a period of 4 s . The minimum coefficient of friction between a heavy body placed on the board if the body does not slip will be

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Options:

A. $\mu = 0.05$

B. $\mu = 0.075$

C. $\mu = 0.173$

D. $\mu = 1.14$

Answer: B

Solution:

To find the minimum coefficient of friction (μ) required for a heavy body to remain stationary on a horizontally oscillating board, we start with the given amplitude and period:

Amplitude (A) = 0.3 m

Period (T) = 4 s

The force associated with an oscillating system is given by $F = kx$. For the body not to slip due to the board's oscillations, the frictional force (f) must be equal to the force due to the oscillation. Thus, we have:

$$f = \mu N = \mu mg$$

Replacing the spring force expression with terms for mass (m) and angular frequency (ω), we have:

$$\mu mg = m\omega^2 A$$

Simplifying this expression to solve for μ :

$$\mu = \frac{\omega^2 A}{g}$$

Knowing that the angular frequency ω can be expressed in terms of the period:

$$\omega = \frac{2\pi}{T}$$

We substitute ω to find μ :

$$\mu = \left(\frac{2\pi}{T}\right)^2 \frac{A}{g}$$

Substituting the provided values:

$$\mu = \left(\frac{2\pi}{4}\right)^2 \frac{0.3}{9.8}$$

Calculate μ :

$$\mu = \frac{4 \times \pi^2 \times 0.3}{4 \times 4 \times 9.8}$$

Finally, compute the numerical value:

$$\mu = 0.075$$

Thus, the minimum coefficient of friction required is $\mu = 0.075$.

Question30

A test tube of mass 6 g and uniform area of cross-section 10 cm^2 is floating in water vertically when 10 g of mercury is in the bottom. The tube is depressed by a small amount and then released. The time period of oscillation is (acceleration due to gravity = 10 ms^{-2})

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Options:

A. 0.75 s

B. 0.5 s

C. 0.25 s

D. 0.85 s

Answer: C

Solution:

The system consists of a test tube and mercury. The total mass is calculated as:

$$\text{Total mass (tube + mercury)} = 6 \text{ g} + 10 \text{ g} = 16 \text{ g} = 0.016 \text{ kg}$$

The area of the test tube's cross-section is given as:

$$A = 10 \text{ cm}^2 = 10^{-3} \text{ m}^2$$

The density of water is:

$$\rho = 10^3 \text{ kg/m}^3$$

The effective spring constant k due to buoyancy is determined using the formula:

$$k = A \times \rho \times g$$

Substituting the known values:

$$k = 10^{-3} \times 10^3 \times 10 = 10$$

Thus, the spring constant k is 10.

The time period of oscillation T is given by:

$$T = 2\pi\sqrt{\frac{m}{k}}$$

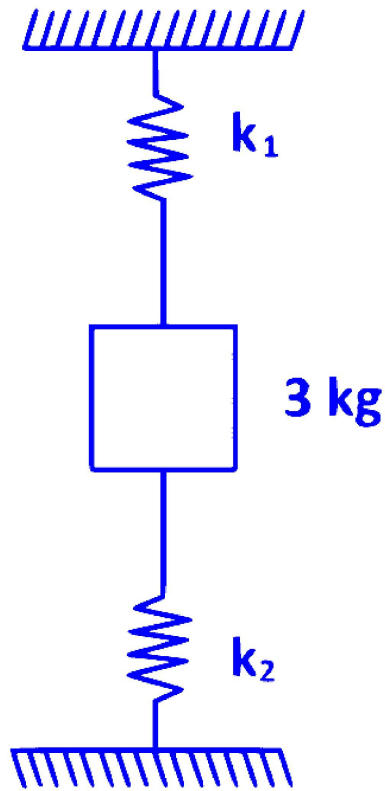
Substituting the mass $m = 0.016 \text{ kg}$ and spring constant $k = 10$:

$$T = 2\pi\sqrt{\frac{0.016}{10}} = 0.25 \text{ s}$$

Therefore, the time period of oscillation is 0.25 seconds.

Question31

A 3 kg block is connected as shown in the figure. Spring constants of two springs k_1 and k_2 are 50Nm^{-1} and 150Nm^{-1} respectively. The block is released from rest with the springs unstretched. The acceleration of the block in its lowest position is $\left(g = 10 \text{ ms}^{-2}\right)$



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Options:

- A. 10 ms^{-2}
- B. 12 ms^{-2}
- C. 8 ms^{-2}
- D. 8.8 ms^{-2}

Answer: A

Solution:

To determine the acceleration of the block at its lowest position, let's start by examining the system's mechanics.

1. Maximum Displacement Calculation

Let x be the maximum downward displacement of the block. The energy conservation equation, considering the force exerted by the springs and gravitational potential energy, is:

$$mgx = \frac{1}{2}(k_1 + k_2)x^2$$

Substitute the known values:

$$x = \frac{2mg}{k_1 + k_2} = \frac{2 \times 3 \times 10}{50 + 150} = \frac{60}{200} = 0.3 \text{ m}$$

2. Acceleration Calculation at Maximum Displacement

The acceleration of the block at this position, due to the net force from the springs and gravity, is given by:

$$a = \frac{(k_1 + k_2)x - mg}{m}$$

Plug in the calculated values:

$$a = \frac{200 \times 0.3 - 3 \times 10}{3} = \frac{60 - 30}{3} = \frac{30}{3} = 10 \text{ m/s}^2$$

Thus, the acceleration of the block at its lowest point is 10 m/s^2 .

Question32

In a time of 2 s , the amplitude of a damped oscillator becomes $\frac{1}{e}$ times, its initial amplitude A . In the next two second, the amplitude of the oscillator is

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Options:

A. $\frac{1}{2\theta}$

B. $\frac{2}{e}$

C. $\frac{1}{e^2}$

D. $\frac{2}{e^2}$

Answer: C

Solution:

The amplitude of a damped oscillator decreases over time due to the damping effect. The relationship for the amplitude is given by:

$$A' = Ae^{-\frac{bt}{2m}}$$

where A is the initial amplitude, b is the damping coefficient, t is the time, and m is the mass of the oscillator.

Initially, at $t = 2$ seconds, the amplitude becomes $\frac{1}{e}$ of the initial amplitude:

$$\frac{A}{e} = Ae^{-\frac{b \times 2}{2m}}$$

This simplifies to:

$$\frac{A}{e} = \frac{A}{e^{\frac{b}{m}}}$$

From this, we deduce:

$$\frac{b}{m} = 1$$

At $t = 4$ seconds (2 more seconds), the amplitude further becomes:

$$A' = Ae^{-\frac{b \times 4}{2m}} = \frac{A}{e^{\frac{4b}{2m}}} = \frac{A}{e^2}$$

Thus, the amplitude at $t = 4$ seconds is proportional to $\frac{1}{e^2}$.

Question33

A particle is executing simple harmonic motion with a time period of 3 s . At a position where the displacement of the particle is 60% of its amplitude. The ratio of the kinetic and potential energies of the particle is

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Options:

A. 5 : 3

B. 16 : 9

C. 4 : 3

D. 25 : 9

Answer: B

Solution:

Given that the time period of the particle executing simple harmonic motion (SHM) is $T = 3$ s.

From this, we know:

$$T = \frac{2\pi}{\omega} \Rightarrow \omega = \frac{2\pi}{3}$$

Let A represent the maximum amplitude. The displacement of the particle is 60% of its amplitude, which means:

$$y = 0.6A$$

For SHM, the kinetic energy (KE) is given by:

$$\text{KE} = \frac{1}{2}m\omega^2 (A^2 - y^2)$$

The potential energy (PE) is:

$$\text{PE} = \frac{1}{2}m\omega^2 y^2$$

The ratio of kinetic energy to potential energy is:

$$\frac{\text{KE}}{\text{PE}} = \frac{\frac{1}{2}m\omega^2 (A^2 - y^2)}{\frac{1}{2}m\omega^2 y^2} = \frac{A^2 - y^2}{y^2}$$

Substituting $y = 0.6A$:

$$\begin{aligned}\frac{\text{KE}}{\text{PE}} &= \frac{A^2 - (0.6A)^2}{(0.6A)^2} \\ &= \frac{A^2(1 - 0.36)}{0.36A^2} \\ &= \frac{0.64A^2}{0.36A^2} \\ &= \frac{0.64}{0.36} \\ &= \frac{16}{9}\end{aligned}$$

Thus, the ratio of the kinetic to potential energy when the displacement is 60% of the amplitude is $\frac{16}{9}$.

Question34

The displacement of a particle executing simple harmonic motion is $y = A \sin(2t + \phi)$ m, where t is time in second and ϕ is phase angle. At time $t = 0$, the displacement and velocity of the particle are 2 m

and 4 ms^{-1} . The phase angle, $\phi =$

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Options:

A. 60°

B. 30°

C. 45°

D. 90°

Answer: C

Solution:

The displacement of a particle undergoing simple harmonic motion is given by $y = A \sin(2t + \phi)$, where t represents time in seconds, and ϕ is the phase angle. At $t = 0$, the displacement is 2 m, and the velocity is 4 ms^{-1} .

Displacement Equation:

$$2 = A \sin(2t + \phi) \quad (\text{Equation 1})$$

Velocity Equation:

The velocity v is the derivative of displacement with respect to time:

$$v = \frac{dy}{dt} = 2A \cos(2t + \phi)$$

At $t = 0$:

$$4 = 2A \cos(2t + \phi)$$

$$\Rightarrow 2 = A \cos(2t + \phi) \quad (\text{Equation 2})$$

Equating Equations (1) and (2):

$$A \sin(2t + \phi) = A \cos(2t + \phi)$$

Simplifying gives:

$$\sin(2t + \phi) = \cos(2t + \phi)$$

$$\sin(2t + \phi) = \sin[90^\circ - (2t + \phi)]$$

This implies:

$$2t + \phi = 90^\circ - (2t + \phi)$$

$$4t + 2\phi = 90^\circ$$

$$2t + \phi = 45^\circ$$

When $t = 0$:

$$\phi = 45^\circ$$

Therefore, the phase angle ϕ is 45° .

Question 35

The displacement of a damped oscillator is $x(t) = \exp(-0.2t) \cos(3.2t + \phi)$, where t is time in second. The time requirement for the amplitude of the oscillator to become $\frac{1}{e^{1.2}}$ times its initial amplitude is

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Options:

A. 3 s

B. 6 s

C. 2 s

D. 8 s

Answer: B

Solution:

To find the time required for the amplitude of a damped oscillator to become $\frac{1}{e^{1.2}}$ times its initial value, consider the displacement of the oscillator given by:

$$x(t) = \exp(-0.2t) \cos(3.2t + \phi)$$

Amplitude Analysis

The amplitude of the oscillator is represented as:

$$A = e^{-0.2t}$$

Initial Amplitude at $t = 0$:

At $t = 0$, the amplitude A_1 is:

$$A_1 = e^{-0.2 \times 0} = 1$$

Amplitude at $t = t'$:

At time $t = t'$, the amplitude A_2 should be:

$$A_2 = \frac{1}{e^{1.2}} A_1 = \frac{1}{e^{1.2}} \times 1 = \frac{1}{e^{1.2}}$$

Solving for t'

To find t' such that:

$$\frac{1}{e^{1.2}} = e^{-0.2t}$$

This relation simplifies to:

$$e^{1.2} = e^{0.2t}$$

Thus, we have:

$$1.2 = 0.2t$$

Therefore, solving for t :

$$t = \frac{1.2}{0.2} = 6 \text{ seconds}$$

Hence, the time required for the amplitude to reduce to $\frac{1}{e^{1.2}}$ times its initial value is 6 seconds.

Question36

Time period of a simple pendulum in air is T . If the pendulum is in water and executes SHM. Its time period is t . The value of $\frac{T}{t}$ is. (density of bob is $\frac{5000}{3} \text{ kg m}^{-3}$)

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Options:

A. $\frac{2}{5}$

B. $\sqrt{\frac{2}{5}}$

C. $\frac{5}{2}$

D. $\sqrt{\frac{5}{2}}$

Answer: D

Solution:

Given:

Time period of oscillation of a pendulum in air is T .

Density of the bob, $\rho = \frac{5000}{3} \text{ kg/m}^3$.

Density of water, $\sigma = 1000 \text{ kg/m}^3$.

We need to find the ratio $\frac{T}{t}$, where t is the time period of the pendulum's oscillation in water.

The time period of oscillation of a pendulum in water can be expressed as:

$$t = \frac{T}{\sqrt{1 - \frac{\sigma}{\rho}}}$$

Substitute the given values:

$$t = \frac{T}{\sqrt{1 - \frac{1000 \times 3}{5000}}}$$

This simplifies to:

$$t = \frac{T}{\sqrt{1 - \frac{3000}{5000}}}$$

Simplifying further:

$$t = \frac{T}{\sqrt{1 - 0.6}}$$

$$t = \frac{T}{\sqrt{0.4}}$$

The ratio $\frac{T}{t}$ is:

$$\frac{T}{t} = \sqrt{\frac{1}{0.4}}$$

$$\frac{T}{t} = \sqrt{\frac{5}{2}}$$

Thus, the value of $\frac{T}{t}$ is $\sqrt{\frac{5}{2}}$.

Question37

For a particle executing simple harmonic motion, Match the following statements (conditions) from Column I to statements (shapes of graph) in Columinit

Column I	Column II
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a Velocity-displacement graph ($\omega = 1$)	i Straight line
b Acceleration-displacement graph	ii Sinusoidal
c Acceleration - time graph	iii Circle
d Acceleration - velocity ($\omega \neq 1$)	iv Ellipse

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Options:

A. a-N, b-i, c-il, d-iif

B. a – in, b – i, c – i, d = k

C. $a = ii, b = 1l, c = 1, d = N$

D. $a = N, b = \bar{c}C = (d - \#$

Answer: B

Solution:

For a body performing simple harmonic motion, the relation between velocity and displacement

$$v = \omega \sqrt{A^2 - x^2}$$

Square both side

$$v^2 = \omega^2 (A^2 - x^2)$$

For ($\omega = 1$)

$$v^2 + x^2 = A^2$$

(a) So, the graph between velocity and displacement will be a circle.

(b) We know that,

$$x = A \sin \omega t \quad \dots (i)$$

Differentiate w.r.t. to t

$$\frac{dx}{dt} = v = +\omega A \cos \omega t \quad \dots (ii)$$

Again differentiate w.r.t. to t ,

$$\frac{d^2x}{dt^2} = a = -\omega^2 A \sin \omega t \quad \dots (iii)$$

From Eq. (i),

$$a = -\omega^2 x \quad \dots \text{(iv)}$$

Comparing from Eq. $y = mx + c$

It is a equation of straight line with slope, $m = -\omega^2$.

So, the graph between acceleration and displacement will be a straight line.

(c) From Eq. (iii),

$$\frac{d^2x}{dt^2} = a = -\omega^2 A \sin \omega t$$

The graph between acceleration and time will be a function of sine wave.

(d) From Eq. (A),

$$v = \omega \sqrt{A^2 - x^2}$$

$$v^2 = \omega^2 (A^2 - x^2)$$

$$x^2 = \left(-\frac{v^2}{\omega^2} + A^2 \right)$$

$$x = \sqrt{A^2 - \frac{v^2}{\omega^2}}$$

Putting this value in Eq. (iv),

$$a = -\omega^2 \sqrt{A^2 - \frac{v^2}{\omega^2}}$$

Square both side

$$a^2 = \omega^4 \left(A^2 - \frac{v^2}{\omega^2} \right)$$

$$a^2 = \omega^4 A^2 - v^2 \omega^2$$

$$\left(\frac{a}{\omega^2} \right)^2 = A^2 - \frac{v^2}{\omega^2}$$

$$\left(\frac{a}{\omega^2} \right)^2 + \left(\frac{v}{\omega} \right)^2 = A^2$$

When $\omega \neq 1$, the graph between acceleration a as a function of velocity v in SHM is an ellipse.

Question38

A particle is executing simple harmonic motion with an instantaneous displacement $x = A \sin^2 \left(\omega t - \frac{\pi}{4} \right)$. The time period of oscillation of the particle is

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Options:

A. $\frac{2\pi}{\omega}$

B. $\frac{\pi}{\omega}$

C. $\frac{\pi}{2\omega}$

D. $\frac{\omega}{2\pi}$

Answer: B

Solution:

Given, instantaneous displacement of particle executing SHM,

$$x = A \sin^2(\omega t - 4)$$
$$\Rightarrow x = A \left[\frac{1 - \cos 2(\omega t - 4)}{2} \right]$$
$$\left(\text{Since, } \sin^2 \theta = \frac{1 - \cos 2\theta}{2} \right)$$

$$\Rightarrow x = \frac{A}{2} [1 - \cos(2\omega t - 8)]$$

Here, angular velocity of particle

$$\omega' = 2\omega$$

$$\Rightarrow \frac{2\pi}{T'} = 2 \times \frac{2\pi}{T} \Rightarrow T' = \frac{T}{2}$$

$$= \frac{\frac{2\pi}{\omega}}{2} = \frac{\pi}{\omega}$$

$$\left(\therefore T = \frac{2\pi}{\omega} \right)$$

Question39

If the amplitude of a lightly damped oscillator decreases by 1.5% then the mechanical energy of the oscillator lost in each cycle is

Options:

- A. 1.5%
- B. 0.75%
- C. 6%
- D. 3%

Answer: D

Solution:

The mechanical energy lost in each cycle is proportional to the square of the amplitude A . Therefore, the fractional change in energy can be expressed as:

$$\begin{aligned}\frac{\Delta E}{E} &\simeq \frac{dE}{E} = \frac{dA^2}{A^2} \\ &= \frac{2AdA}{A^2} = 2\frac{dA}{A} \\ &= 2 \times 1.5\% = 3\%\end{aligned}$$

Thus, the mechanical energy of the oscillator lost in each cycle is 3%.

Question40

A body is executing S.H.M. At a displacement x its potential energy is 9 J and at a displacement y its potential energy is 16 J . The potential energy at displacement $(x + y)$ is

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Options:

- A. 25 J
- B. 5 J
- C. 49 J
- D. 7 J

Answer: C

Solution:

According to first condition,

In S. H. M, at displacement x

Potential energy = 9 J

$$\text{i.e., } \frac{1}{2}kx^2 = 9 \quad \dots (i)$$

Again, at displacement y , Potential energy

$$\frac{1}{2}ky^2 = 16 \quad \dots (ii)$$

Dividing Eq. (ii) by Eq. (i), we get

$$\begin{aligned} \frac{\frac{1}{2}ky^2}{\frac{1}{2}kx^2} &= \frac{16}{9} \\ \Rightarrow \left(\frac{y}{x}\right)^2 &= \left(\frac{4}{3}\right)^2 \\ \frac{y}{x} &= \frac{4}{3} \quad \dots (iii) \end{aligned}$$

$\therefore$ Potential energy of body at displacement $(x + y)$

$$\begin{aligned} \text{PE} &= \frac{1}{2}k(x + y)^2 \\ &= \frac{9}{x^2}(x + y)^2 \quad [\text{From Eq. (i)}] \\ &= 9 \frac{(x + y)^2}{x^2} = 9 \left(1 + \frac{y}{x}\right)^2 \\ &= 9 \left(1 + \frac{4}{3}\right)^2 \quad [\text{From Eq. (iii)}] \\ &= 49 \text{ J} \end{aligned}$$

Question41

A hydrometer executes simple harmonic motion when it is pushed down vertically in a liquid of density ρ . If the mass of hydrometer is m and the radius of the hydrometer tube is r , then the time period of oscillation is

Options:

A. $T = 2\pi\sqrt{\frac{m}{\pi^2\rho g}}$

B. $T = 2\pi\sqrt{\frac{\pi\pi^2\rho g}{m}}$

C. $T = \frac{1}{2\pi}\sqrt{\frac{m}{\pi\pi^2\rho g}}$

D. $T = \frac{1}{2\pi}\sqrt{\frac{\pi\pi^2\rho g}{m}}$

Answer: A

Solution:

If the hydrometer is in equilibrium, the weight is balanced by the buoyancy force on it by the fluid. When hydrometer liquid is pressed by x distance the net unbalanced force in the upward direction is given by $F = -(\text{Area} \times \text{Density} \times \text{Gravity} \times \text{Distance moved by fluid})$

Where, $-ve$ sign shows that the direction of force will be opposite to the direction of motion.

Thus, $m\frac{d^2x}{dt^2} = -\pi r^2\rho gx$

Where, r = radius of liquid column

ρ = density of liquid

x = distance moved by liquid

Now, $\frac{d^2x}{dt^2} = \frac{-\pi r^2\rho g}{m}x$

or $\omega^2 = \frac{\pi r^2\rho g}{m}$ or $\omega = \sqrt{\frac{\pi r^2\rho g}{m}}$

$\Rightarrow T = 2\pi\sqrt{\frac{m}{\pi r^2\rho g}}$

Question42

An object undergoing simple harmonic motion takes 0.5 s to travel from one point of zero velocity to the next such point. The angular frequency of the motion is

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Options:

A. $\pi \text{ rad s}^{-1}$

B. $2\pi \text{ rad s}^{-1}$

C. $3\pi \text{ rad s}^{-1}$

D. $\frac{\pi}{2} \text{ rad s}^{-1}$

Answer: B

Solution:

The velocity of the object becomes zero at maximum displacement of either side of the mean position in the SHM.

Time taken from one maximum to another maximum = 0.5 s

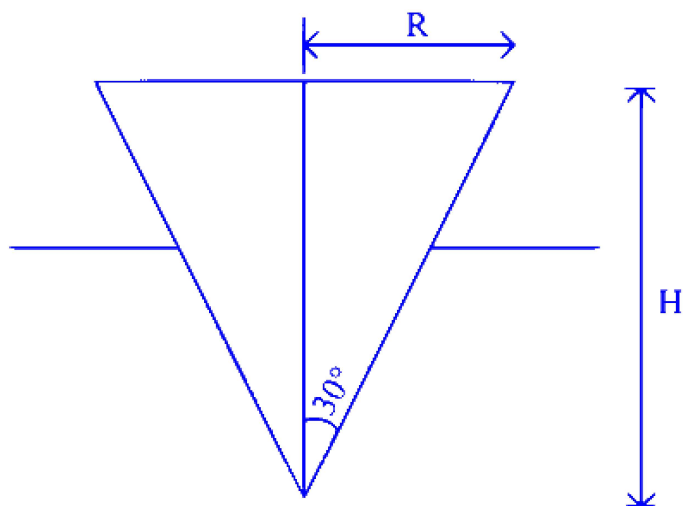
Time period = $2 \times 0.5 = 1 \text{ s}$

Angular velocity, $\omega = 2\pi v$

$$= \frac{2\pi}{T} = \frac{2\pi}{1} = 2\pi \text{ rad/s}$$

Question43

A cone with half the density of water is floating in water as shown in figure. It is depressed down by a small distance $\delta (\ll H)$ and released. The frequency of simple harmonic oscillations of the cone is



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Options:

A. $\frac{1}{2\pi} \sqrt{\frac{6g}{H} \frac{1}{4^{\frac{1}{3}}}}$

B. $\frac{1}{2\pi} \sqrt{\frac{3g}{H} \frac{1}{4^{\frac{1}{3}}}}$

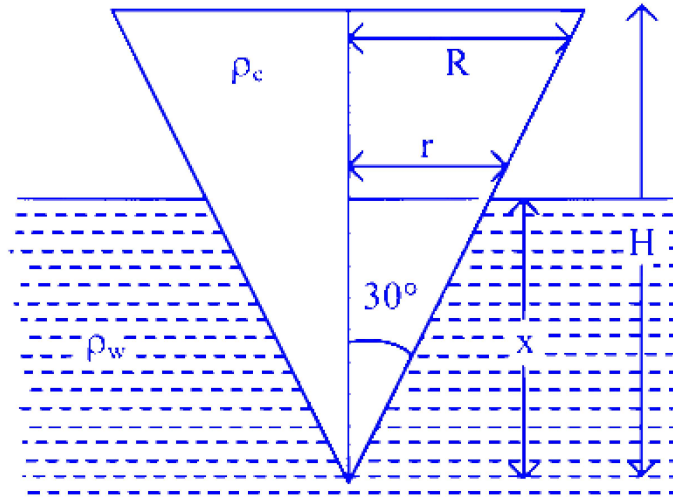
C. $\frac{1}{2\pi} \sqrt{\frac{6g}{2H}}$

D. $\frac{1}{2\pi} \sqrt{\frac{g}{H}}$

Answer: A

Solution:

Let initially the cone be in equilibrium i.e. floating



From the diagram, it is clear that

Buoyant force (F_b) = weight of the cone. (F_w)

$$\rho_w \times \left(\frac{1}{3} \pi r^2 x \right) \times g = \rho_c \times \left(\frac{1}{3} \pi R^2 H \right) \times g$$

$$\Rightarrow 2r^2 x = R^2 H \quad (\because \rho_c = \frac{\rho_w}{2})$$

$$\Rightarrow x = \frac{R^2 H}{2r^2} \quad \dots (i)$$

From the diagram,

$$\tan \theta = \frac{r}{x} = \frac{R}{H}$$

$$\Rightarrow x = \frac{rH}{R} \quad \dots (ii)$$

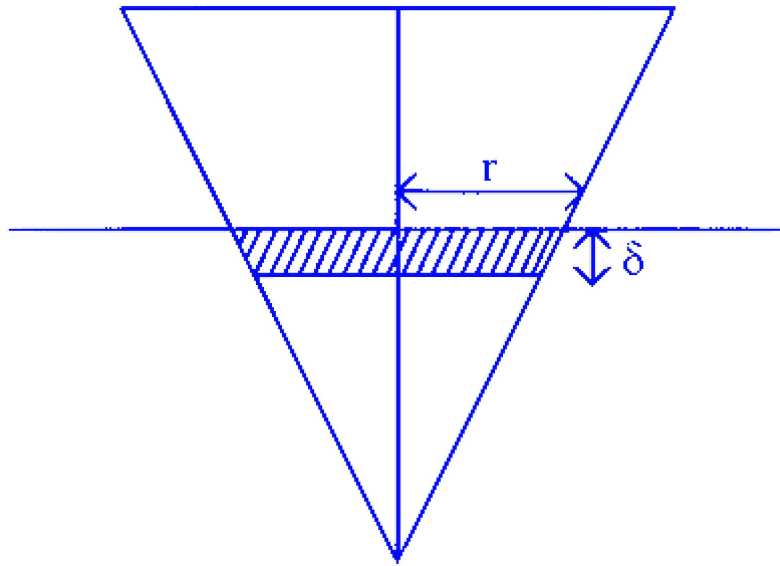
From Eqs. (i) and (ii)

$$\frac{R^2 H}{2r^2} = \frac{rH}{R}$$

$$\Rightarrow \frac{R^3}{2} = r^3 \Rightarrow r = \left(\frac{1}{2} \right)^{1/3} R$$

Now, the cone. is depressed by small distance $\delta (<< H)$ and released it executes simple harmonic motion. In simple harmonic motion, the buoyancy force acts as restoring force which is given as

$$F_b = \text{density} \times \text{volume} \times \text{gravity}$$



Since, the displacement is very small, the excess submerged shape of cone. can be considered as a cylinder whose volume is given by $\pi r^2 \delta$.

$$\therefore F_b = \rho_w \pi r^2 \delta g = \rho_w \pi \left[\left(\frac{1}{2} \right)^{1/3} R \right]^2 \delta g$$

Thus, force is equal to restoring force, hence

$$F_b = k\delta = m\omega^2 \delta$$

$$\rho_w \pi \left(\frac{1}{4} \right)^{1/3} R^2 \delta g = \rho_c \frac{1}{3} \pi R^2 H \delta \omega^2$$

$$2 \times 3 \times \left(\frac{1}{4} \right)^{1/3} \delta g = H \delta \omega^2 \quad (\because \rho_c = \frac{\rho_w}{2})$$

$$\Rightarrow \frac{6 \left(\frac{1}{4} \right)^{1/3} g}{H} = \omega^2$$

$$\text{or } \omega = \sqrt{\frac{6g}{H} \left(\frac{1}{4} \right)^{1/3}} \Rightarrow 2\pi f = \sqrt{\frac{6g}{H} \left(\frac{1}{4} \right)^{1/3}}$$

$$\Rightarrow f = \frac{1}{2\pi} \sqrt{\frac{6g}{H} \left(\frac{1}{4} \right)^{1/3}} = \frac{1}{2\pi} \sqrt{\frac{6g}{H} \cdot \frac{1}{4^{1/3}}}$$

Question44

In case of a forced vibration, the resonance wave becomes very sharp when the

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Options:

A. quality factor is small

B. damping force is small

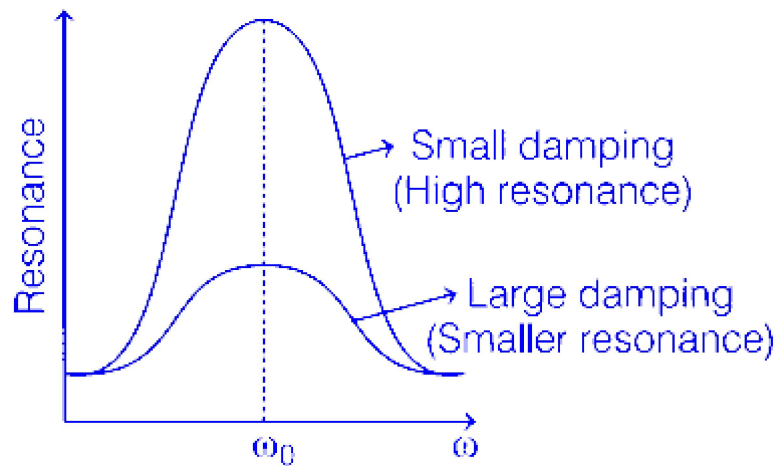
C. restoring force is small

D. applied periodic force is small

Answer: B

Solution:

As we know that, for less damping, resonance peak is taller and sharper.



Question45

A particle executing simple harmonic motion along a straight line with an amplitude A , attains maximum potential energy when its displacement from mean position equals

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Options:

A. 0

B. $\pm \frac{A}{\sqrt{2}}$

C. $\pm A$

D. $\pm \frac{A}{2}$

Answer: C

Solution:

Let, wave displacement be x .

As we know that,

Potential energy, $U = \frac{1}{2}kx^2$

where, k is wave constant.

Potential energy will be maximum, when displacement is maximum.

$$\Rightarrow x = \pm A$$

Question 46

The bob of a simple pendulum is a spherical hollow ball filled with water. A plugged hole near the bottom of the oscillating bob gets suddenly unplugged. During observation, till water is coming out the time period of oscillation would

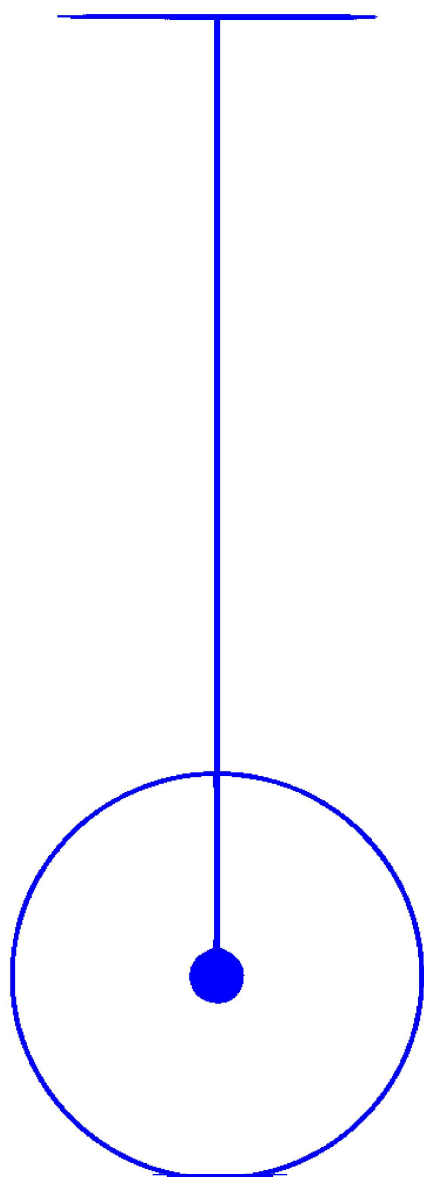
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Options:

- A. remain unchanged
- B. increase towards a saturation value
- C. first increase and then decrease to the original value
- D. first decrease and then increase to the original value

Answer: C

Solution:



Let,

Time period = T

Length of simple pendulum = l

Acceleration due to gravity = g

$$\therefore T = 2\pi\sqrt{l/g} \dots (i)$$

When sphere is fully filled with water, centre of mass lies at centre.

Case I For, water level below centre of sphere

$$l_1 > l_0 \dots (ii)$$

Case II For empty sphere,

$$l = l_0 < l_1 \dots (iii)$$

$\therefore$ From Eqs. (i), (ii) and (iii)

In case I, time period increase.

In case II, time period decrease.

Question47

A block of mass l kg is fastened to a spring of spring constant of 100 Nm^{-1} . The block is pulled to a distance $x = 10 \text{ cm}$ from its equilibrium position ($x = 0 \text{ cm}$) on a frictionless surface, from rest at $t = 0$. The kinetic energy and the potential energy of the block when it is 5 cm away from the mean position is

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Options:

A. 0.375 J , 0.125 J

B. 0.125 J , 0.375 J

C. 0.125 J , 0.125 J

D. 0.375 J , 0.375 J

Answer: A

Solution:

Given, mass of block, $m = 1 \text{ kg}$

Spring constant, $k = 100 \text{ N/m}$

In case I, extension in spring, $\Delta x_1 = 10 \text{ cm}$

$$= 10 \times 10^{-2} \text{ m}$$

Initial speed, $u = 0 \text{ ms}^{-1}$

In case II, extension in spring $\Delta x_2 = 5 \text{ cm}$

$$= 5 \times 10^{-2} \text{ m}$$

Let, kinetic energy and potential energy be (KE) and (PE)

As we know that,

$$\text{KE} = \frac{1}{2}k(\Delta x_1^2 - \Delta x_2^2)$$

$$= \frac{1}{2} \times 100 (10^2 - 5^2) \times 10^{-4}$$

$$= 50 \times 75 \times 10^{-4} = 0.375 \text{ J}$$

$$\text{and PE} = \frac{1}{2}k\Delta x_2^2$$

$$= \frac{1}{2} \times 100 \times (5 \times 10^{-2})^2 = 50 \times 25 \times 10^{-4} = 0.125 \text{ J}$$

Question48

The scale of a spring balance which can measure from 0 to 15 kg is 0.25 m long. If a body suspended from this balance oscillates with a time period $\frac{2\pi}{5} \text{ s}$, neglecting the mass of the spring, find the mass of the body suspended.

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Options:

A. 24 kg

B. 1 kg

C. 20 kg

D. 7 kg

Answer: A

Solution:

Given, range of mass is from m_1 to m_2 i.e. 0 to 15 kg

Length of spring, $x = 0.25$ m

Time period of oscillation, $T = \frac{2\pi}{5}$ s

Let mass of body suspended = m

Spring constant = k

Acceleration due to gravity $g = 10 \text{ ms}^{-2}$

Since, $T = 2\pi\sqrt{\frac{m}{k}}$ (i)

and $F = mg = kx$

$$\Rightarrow k = \frac{mg}{x} = \frac{15 \times 10}{0.25} \text{ N/m}$$

Substituting in Eq. (i), we get

$$\begin{aligned} T &= 2\pi\sqrt{\frac{m \times 25}{15 \times 10 \times 100}} \\ \Rightarrow \left(\frac{2\pi}{5}\right)^2 &= (2\pi)^2 \left(\frac{m}{15 \times 40}\right) \\ \Rightarrow \frac{4\pi^2}{25} &= \frac{4\pi^2 m}{15 \times 40} \Rightarrow m = 24 \text{ kg} \end{aligned}$$

Question49

A spring is stretched by 0.40 m when a mass of 0.6 kg is suspended from it. The period of oscillations of the spring loaded by 255 g and put to oscillations is close to ($g = 10 \text{ ms}^{-2}$)

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Options:

- A. 11 s
- B. 48.6 s
- C. 0.82 s
- D. 4.86 s

Answer: C

Solution:

Given, expansion of spring, $\Delta x = 0.4 \text{ m}$

Mass suspended in case I (m_1) and case II (m_2) is 0.6 kg and 255 g, respectively.

Acceleration due to gravity $g = 10 \text{ ms}^{-2}$

Since, force $F = mg = k\Delta x$

$$\therefore \text{Spring constant } k = \frac{mg}{\Delta x}$$

$$\Rightarrow k = \frac{m_1 g}{\Delta x_1} = \frac{0.6 \times 10}{0.4} = 15 \text{ N/m}$$

and time period, $T = 2\pi\sqrt{\frac{m}{k}}$

$$T_2 = 2\pi\sqrt{\frac{m_2}{k}}$$

$$T_2 = 2\pi\sqrt{\frac{0.255}{15}}$$

$$= 2\pi\sqrt{\frac{255}{15 \times 10^3}}$$

$$= 2\pi\sqrt{0.017} = 0.82 \text{ s}$$

Question 50

A heavy brass sphere is hung from a spring and it executes vertical vibrations with period T . The sphere is now immersed in a non-viscous liquid with a density $(1/10)$ th that of brass. When set into vertical vibrations with the sphere remaining inside the liquid all the time, the time period will be

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Options:

A. $\left(\sqrt{\frac{9}{10}}\right)T$

B. $\left(\sqrt{\frac{10}{9}}\right)T$

C. $\left(\frac{9}{10}\right)T$

D. Unchanged

Answer: D

Solution:

Given, time period of vibration in air = T

Density of fluid = $\rho/10$

Time period of vibration in fluid = T'

Since, $T = 2\pi\sqrt{\frac{m}{k}}$

and as the value of mass (m) and spring constant(k) of spring-mass system is same.

$\therefore$ Time period will remain same.
